



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Tiny AI chip with voice detection as well as extraction

POLYN neuromorphic analogue ultra-low-power chip is the industry's first solution to combine voice detection and voice extraction in a single compact package. The tiny AI chip is capable of extracting voice from any background noise, thus enabling hearables such as wireless earphones, hearing aids, gamer headphones, and intercom systems to provide better sound quality even in noisy environments. The system uses an AI based extraction of the voice, which provides better hearing experience than standard noise cancellation filters. It offers especially clear voice and immediate adaptation to any background conditions along with voice extraction/transparent mode. It can enable security personnel and fire fighters to communicate better during rescue operations in noisy environments.

POLYN Technology
<https://polyn.ai>

Sensor with long-range detection of moving objects

The Eyeonic Vision sensor from SiLC Technologies can identify objects at distances greater than a kilometre.



This sensor integrates millimetre-level accuracy, depth, and instantaneous velocity, thus enabling the sensor to detect distance, speed, and direction of any moving object instantly. The

sensor can enable new capabilities for various machine vision applications, such as autonomous cars and drones. It can enable long-distance detection of objects, which can improve safety and enable faster path planning. Ultra-long-range visibility finds usage in multiple industries that utilise machine vision, including automotive, metrology, and construction.

SiLC Technologies, Inc.
<https://www.silc.com>

SiP with BLE and LoRa lets IoT devices work for years

The iMCP HTLRBL32L system-in-package (SiP) from HT Micron can enable communication to an IoT solution. The SiP combines Bluetooth Low Energy (BLE) and LoRa to enable easy integration and prototyping



of long- and short-range IoT solutions. The product comes in a small package measuring just 13mm × 13mm × 1.1mm. The low-power solution can enable IoT devices to work for years on a single battery charge. It features an accessible micro-controller and internal memory, and a full range of interfaces for analogue and digital peripherals. The SiP also provides high configurability and flexibility for IoT applications. It can enable designers to build low-cost, low-power, compact IoT solutions that can find use in a large field of applications, including, smart farming, smart building, and smart logistics.

HT Micron
<http://www.htmicron.com.br>

World's first all-in-one full-colour microLED display

Porotech has announced the world's first all-in-one full-colour microLED display. The display uses Porotech's DynamicPixelTuning (DPT) technology that enables



a single pixel to emit light across the entire colour spectrum. The display employs a class of gallium nitride (GaN) materials, which enables

a single microLED diode to emit all colours. The current technology employs RGB (red-green-blue) sub-pixel model, which increases the size of the display, reduces resolution and cost of manufacturing. The DPT can enhance the pixel density and resolution by up to four times, thus lowering the cost of displays, reducing power consumption, and providing better colour and depth of image. This technology can be used in applications such as AR and VR headsets.

Porotech
<https://www.porotech.com>

The smallest laser spectrometer chip for health monitoring

Rockley Photonics has released the world's smallest laser spectrometer chip for health monitoring applications. The laser based solution has a smaller area than the LED based spectrometer. It is the world's first micro-transfer-printed (mTP) silicon-photonics based laser for commercial applications. It offers up to 6x higher density than any other existing product, which can enable engineers to create smaller wearable devices. The IC offers photonic chip size reduction and